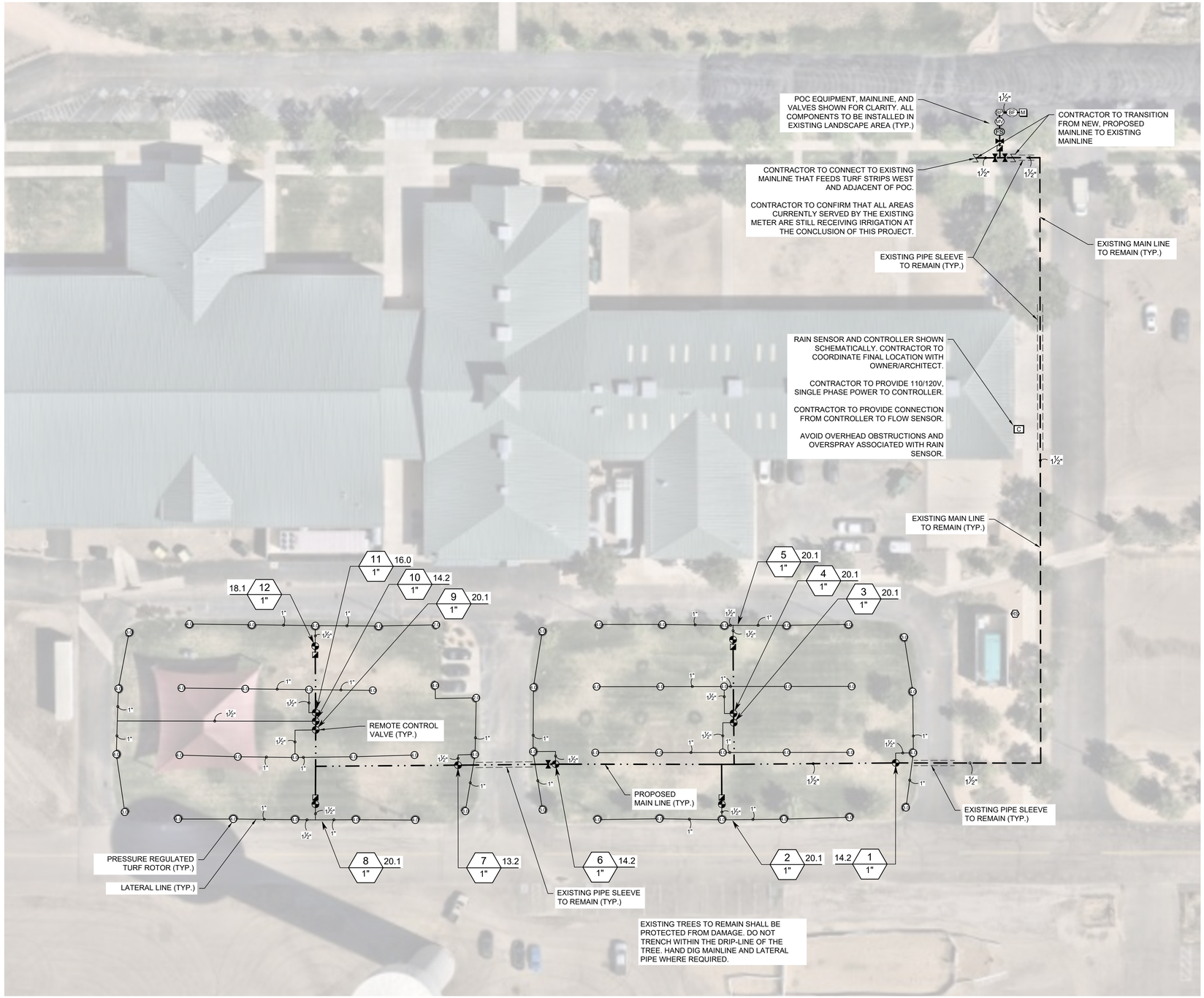


IRRIGATION SCHEDULE

SYMBOL	MANUFACTURER/MODEL/DESCRIPTION	QTY	PSI
2.0	RAIN BIRD 5006-PL-PC-SAM-R 2.0 TURF ROTOR, 6IN. POP-UP WITH PLASTIC RISER. ADJUSTABLE TO FULL CIRCLE. STANDARD ANGLE NOZZLE, IN-STEM PRESSURE REGULATOR, AND FLOW SHUT-OFF DEVICE. WITH SEAL-A-MATIC CHECK VALVE.	2	45
3.0	RAIN BIRD 5006-PL-PC-SAM-R 3.0 TURF ROTOR, 6IN. POP-UP WITH PLASTIC RISER. ADJUSTABLE TO FULL CIRCLE. STANDARD ANGLE NOZZLE, IN-STEM PRESSURE REGULATOR, AND FLOW SHUT-OFF DEVICE. WITH SEAL-A-MATIC CHECK VALVE.	7	45
4.0	RAIN BIRD 5006-PL-PC-SAM-R 4.0 TURF ROTOR, 6IN. POP-UP WITH PLASTIC RISER. ADJUSTABLE TO FULL CIRCLE. STANDARD ANGLE NOZZLE, IN-STEM PRESSURE REGULATOR, AND FLOW SHUT-OFF DEVICE. WITH SEAL-A-MATIC CHECK VALVE.	46	45

SYMBOL	MANUFACTURER/MODEL/DESCRIPTION	QTY
	IRRITROL 2400T 1" ELECTRIC REMOTE CONTROL VALVE, 1IN., GLOBE CONFIGURATION, NPT THREADS	12
	RAIN BIRD 44-LRC 1" 1IN. BRASS QUICK-COUPLING VALVE, WITH CORROSION-RESISTANT STAINLESS STEEL SPRING, LOCKING THERMOPLASTIC RUBBER COVER, AND 2-PIECE BODY.	5
	BRASS ISOLATION BALL VALVE	4
MV	IRRITROL 700 WITH OMR-100 AND SPK-100 1-1/2" ELECTRIC MASTER VALVE, WITH OMNI-REG 7PSI-100PSI REGULATOR, WEATHERPROOF GAUGE	1
BF	FEBCO 825YA 1" REDUCED PRESSURE BACKFLOW PREVENTER	1
C	HYDRO POINT WTOXR-C-18-CWM 18-GAUGE, POWDER-COATED, COLD-ROLLED STEEL, WALL MOUNT ENCLOSURE W/ KEY-LOCK ENTRY.	1
RS	IRRITROL RFS1000 WIRELESS RAIN/FREEZE SENSOR. MOUNT AS NOTED OR APPROVED, USE CONTROLLER POWER OR OPTIONAL TRANSFORMER.	1
FS	HYDRO POINT WTFLOWHD-P-150 1.5" PVC, HIGH-DEFINITION FLOW SENSOR AND SUB-METER FOR 1.5IN. MAINLINES.	1
BP	PRECISION NOVA BOOSTER PUMP, 208V / 3 PHASE MODEL # CB##V1C002X00025-040XXXX203ONS-2-SS CONTACT ROBERT DICKSON (208) 323-5300 208V / 3 PHASE POWER HAS BEEN CONFIRMED ON-SITE.	1
M	WATER METER 1" EXISTING WATER METER TO REMAIN. CONTRACTOR TO REPLACE EXISTING BACKFLOW WITH RPZ PER PLANS. 55 PSI CONFIRMED AT EXISTING BACKFLOW	1
	IRRIGATION LATERAL LINE: PVC SCHEDULE 40 1"	1,507 LF
	IRRIGATION LATERAL LINE: PVC SCHEDULE 40 1 1/2"	319.3 LF
	IRRIGATION MAINLINE: PVC SCHEDULE 40 1 1/2" EXISTING 1.5" MAINLINE CONTRACTOR TO NOTIFY OWNER AND IRRIGATION DESIGNER IF MAINLINE IS SMALLER THAN 1.5".	506.7 LF
	IRRIGATION MAINLINE: PVC SCHEDULE 40 1 1/2" NEW, PROPOSED MAINLINE TO BE INSTALLED BY CONTRACTOR PER PLANS	518.1 LF
	PIPE SLEEVE: PVC SCHEDULE 40	147.8 LF

Valve Callout
- Valve Number
- Valve Flow
- Valve Size

NOTE: CONTRACTOR TO REMOVE ALL EXISTING COMPONENTS IN THE SOUTH LAWNS TURF AREAS. EXAMPLES OF COMPONENTS ARE EXISTING VALVES, VALVE BOXES, AND HEADS.

EXISTING PIPE WILL BE ABANDONED IN PLACE.

NORTH

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0 15 30 60

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PAUL REED
Certified Irrigation Designer
CID
84393

KHA PROJECT	067606016
DATE	7/1/2025
SCALE	AS SHOWN
DESIGNED BY	PMR
DRAWN BY	PMR
CHECKED BY	AMC

IRRIGATION PLANS

ARAPAHOE COUNTY FAIRGROUNDS
25690 E QUINCY AVE
AURORA, CO 80016

SHEET NUMBER
IR 1.0

Plotted By:Reed, Paul Sheet Set:Kha Layout:IR 1.0 IRRIGATION PLANS July 11, 2025 03:23:52pm K:\DEN_LA\067606016 — Arapahoe County Fairgrounds\CAD\PlanSheets\067606016_IRR.dwg
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GENERAL IRRIGATION SPECIFICATIONS AND NOTES

1. PROJECT DESIGNED WITH A "STREET" STATIC PRESSURE OF 55 PSI. BOOSTER PUMP TO BOOST PRESSURE BY 30 PSI.
   CONTRACTOR TO VERIFY ACTUAL AVAILABLE WATER PRESSURE BEFORE BEGINNING INSTALLATION. CONTRACTOR SHALL NOTIFY LANDSCAPE ARCHITECT IF AVAILABLE WATER PRESSURE IS LOWER THAN THE AS-DESIGNED PRESSURE FOUND IN THE CRITICAL ANALYSIS. CRITICAL ANALYSIS FOR EACH POC / METER CAN BE FOUND IN THIS PLAN SET.
2. THE SYSTEM SHALL BE DESIGNED TO PROVIDE 100% COVERAGE. ANY CHANGES MADE IN THE LAYOUT DUE TO FIELD CONDITIONS SHALL BE IN ACCORDANCE WITH THESE STANDARDS. CONTRACTOR SHALL NOTIFY LANDSCAPE ARCHITECT OF ANY DISCREPANCIES IN PLANS OR SPECIFICATIONS PRIOR TO BEGINNING OR CONTINUING WORK. THIS PLAN IS SCHEMATIC AND DUE TO THE NATURE OF CONSTRUCTION SLIGHT FIELD MODIFICATIONS MAY BE NECESSARY TO IMPLEMENT PLAN.
3. THE CONTRACTOR SHALL MAKE NO SUBSTITUTIONS, DELETIONS, OR ADDITIONS TO THIS PLAN WITHOUT APPROVAL OF THE LANDSCAPE ARCHITECT.
4. ALL CONSTRUCTION SHALL CONFORM TO TOWN, COUNTY, STATE, AND FEDERAL REQUIREMENTS. IT SHALL BE THE RESPONSIBILITY OF THE IRRIGATION CONTRACTOR TO ENSURE THAT ALL IRRIGATION EQUIPMENT MEETS GOVERNMENT REGULATIONS. CONTRACTOR SHALL ALSO BE RESPONSIBLE FOR OBTAINING ANY NECESSARY PERMITS OR APPROVALS.
5. VERIFY LOCATIONS OF ALL UNDERGROUND UTILITIES PRIOR TO INSTALLATION OF IRRIGATION SYSTEM. ALL UTILITIES AND STRUCTURES MAY NOT BE SHOWN ON THESE PLANS-CONTRACTOR SHALL FIELD VERIFY.
6. CONTRACTOR TO FIELD VERIFY ALL POINT OF CONNECTION SOURCE INFORMATION PRIOR TO CONSTRUCTION.
7. IRRIGATION DESIGN IS SCHEMATIC ONLY. FULL AND COMPLETE SHOP DRAWINGS SHALL BE SUBMITTED FOR REVIEW BY THE OWNER'S REPRESENTATIVE.
8. CONTRACTOR SHALL COORDINATE WITH THE PLANTING PLAN FOR PLANTER BED AND TREE LOCATIONS TO ENSURE ALL PLANT MATERIAL IS COVERED BY 100% HEAD-TO-HEAD IRRIGATION.
9. CONTRACTOR SHALL PROVIDE "AS-BUILT" DRAWINGS OF THE FINAL INSTALLATION TO OWNER AT SUBSTANTIAL COMPLETION BEFORE RECEIVING FINAL PAYMENT.
10. IRRIGATION CONTRACTOR TO COORDINATE POWER SUPPLY TO BOOSTER PUMP AND IRRIGATION CONTROLLER WITH ELECTRICAL CONTRACTOR.
11. IRRIGATION CONTRACTOR SHALL SECURE ANY AND ALL NECESSARY PERMITS FOR THE WORK PRIOR TO COMMENCEMENT OF OPERATIONS ON-SITE. COPIES OF THE PERMITS SHALL BE SENT TO THE OWNER/GENERAL CONTRACTOR. WORK IN THE R.O.W. SHALL CONFORM TO THE STANDARDS AND SPECIFICATIONS OF LOCAL AND/OR STATE JURISDICTION.
12. IRRIGATION LATERAL LINES, MAIN LINES AND EQUIPMENT MAY BE SHOWN OUTSIDE PROPERTY LINES ON THIS PLAN, ALL IRRIGATION LINES AND EQUIPMENT ARE TO BE WITHIN AND INSTALLED WITHIN THE LIMITS OF THE PROPERTY LINE.
13. LOCATE ALL IRRIGATION LINES WITHIN LANDSCAPED AREAS WHENEVER POSSIBLE. ALL LINES UNDER PAVEMENT MUST BE SLEEVED. ALL VALVES SHALL BE LOCATED WITHIN LANDSCAPED AREAS, AS SPECIFIED.
12. IRRIGATION SYSTEMS CONNECTED TO A POTABLE WATER SUPPLY SHALL HAVE A REDUCED PRESSURE ZONE BACKFLOW PREVENTER INSTALLED.
13. SUPPLY LINE AND METER TO BE PROVIDED BY GENERAL CONTRACTOR. BACKFLOW PREVENTER TO BE PROVIDED BY IRRIGATION CONTRACTOR. IRRIGATION CONTRACTOR'S POINT OF CONNECTION TO BEGIN AFTER THE IRRIGATION WATER METER.
14. INSTALL ALL BACKFLOW PREVENTION DEVICES AND ALL PIPING BETWEEN THE POINT OF CONNECTION AND THE BACKFLOW PREVENTER AS PER LOCAL CODES. FINAL LOCATION SHALL BE DETERMINED BY THE OWNER'S AUTHORIZED REPRESENTATIVE.
15. CONTRACTOR TO COORDINATE LOCATION OF ALL METERS AND BACKFLOW ASSEMBLIES WITH PROJECT OWNER.
16. THE IRRIGATION CONTRACTOR SHALL BE DIRECTLY RESPONSIBLE FOR SLEEVING AND DIRECTIONAL BORES.
17. EXISTING TREES TO REMAIN ARE TO BE PROTECTED FROM DAMAGE. DO NOT TRENCH OR EXCAVATE WITHIN THE CRITICAL ROOT ZONE OF ANY TREE. EXAMPLE: A 12" CALIPER TREE HAS A CRITICAL ROOT ZONE THAT EXTENDS 12 FEET FROM THE BASE OF THE TRUNK.
18. ALL SLEEVES UTILIZED BY THE IRRIGATION CONTRACTOR WHETHER INSTALLED BY THE IRRIGATION CONTRACTOR OR NOT, SHALL BE LOCATED ON THE "AS-BUILT" DRAWINGS. THE DEPTH BELOW FINISH GRADE, TO THE NEAREST FOOT OF EACH END OF THE SLEEVE SHALL BE NOTED AT EACH SLEEVE LOCATION ON THE "AS-BUILT" DRAWINGS. ALL SLEEVES ON PLAN FOR WALL PENETRATIONS AND UNDER SIDEWALKS SHALL BE SIZED TWO PIPE SIZES GREATER THAN THE PIPE IT CARRIES. ALL IRRIGATION SLEEVING TO BE THE RESPONSIBILITY OF THE IRRIGATION CONTRACTOR. ELECTRICAL WIRES FOR IRRIGATION VALVES AND IRRIGATION LINES ARE TO BE PLACED IN SEPARATE SLEEVES. SEE SLEEVING DETAIL.
19. ALL PRESSURIZED MAINLINES AND LATERALS UNDER PAVEMENT SHALL BE WITHIN SLEEVES AS NOTED. WHERE ELECTRIC OR HYDRAULIC VALVE CONTROL LINES PASS THROUGH A SLEEVE WITH OTHER MAIN OR LATERAL LINES THEY SHALL BE CONTAINED WITHIN A SEPARATE, SMALLER CONDUIT.
20. SLEEVES UNDER EXISTING PAVEMENT MUST BE DIRECTIONAL BORE. OPEN CUT IS NOT AN OPTION.
21. NUMBER THE TOP OF ALL VALVE BOX LIDS WITH MINIMUM 1" HEIGHT BLACK LETTERS TO CORRESPOND TO AUTOMATIC AND GATE VALVE DESIGNATIONS. ALL QUICK COUPLER VALVE BOXES SHALL BE LABELED IN A SIMILAR MANNER WITH THE DESIGNATION "QC". LETTER OUTSIDE OF TIME CLOCK CABINETS TO CORRESPOND WITH IRRIGATION CLOCK PROGRAM DESIGNATION.
22. THE IRRIGATION CONTRACTOR SHALL INSTALL A COLOR CODED METAL DETECTABLE MARKING TAPE WHICH CLEARLY NOTES: "CAUTION: IRRIGATION LINE BURIED BELOW," THE TAPE SHALL BE INSTALLED THE FULL LENGTH OF THE IRRIGATION MAINLINE.
23. ALL VALVES AND WIRE SPLICES SHALL BE LOCATED WITHIN VALVE BOXES. WIRE SPLICE KITS TO BE RATED "DIRECT BURIAL" BUT STILL NEED TO BE ABOVE GROUND (INSIDE VALVE BOX).
24. ALL UNSIZED PIPE SHALL BE 1" UNLESS OTHERWISE NOTED ON PLAN. SEE LATERAL PIPE SIZING REQUIREMENTS.
25. LOCATE ALL VALVES IN PLANTING BEDS WITH A MINIMUM OF 3'-0" FROM BACK OF CURB OR EDGE OF PAVEMENT, UNLESS OTHERWISE NOTED. PIPE SIZES ON EITHER SIDE OF SECTION VALVES CONNECTING MAINLINE TO SECTION LATERAL SHALL BE ONE (1) PIPE SIZE LARGER THAN VALVE SIZE. WHERE MAINLINES RUN PARALLEL TO PAVEMENT OR CURBING, THE MAINLINE SHALL BE OFFSET 2'-0" FROM THE EDGE OF PAVEMENT OR CURB.
26. IRRIGATION ZONES SHALL BE SEPARATED FOR HIGH AND LOW WATER USE REQUIREMENTS AND OPERATED ON DIFFERENT WATERING CYCLES. BUBBLERS, DRIPLINE, AND SPRAY HEADS SHALL BE SEPARATED ON DIFFERENT VALVES.
27. ALL DRIP ZONES SHALL BE INSTALLED WITH A FLUSH VALVE AND DRIP INDICATOR.
28. IRRIGATION CONTRACTOR TO COORDINATE WITH OWNER FOR FINAL CONTROLLER AND RAIN SENSOR LOCATIONS. THE CONTROLLER SHALL BE PLACED IN A LOCKING CABINET APPROPRIATE FOR ITS LOCATION. (INDOOR VS. OUTDOOR USE)
29. IRRIGATION CONTRACTOR SHALL REVIEW WINTERIZATION PROCEDURES FOR IRRIGATION SYSTEM WITH OWNERS REPRESENTATIVE.
30. LOCATE THE AUTOMATIC RAIN / FREEZE SENSOR SHUTOFF DEVICE IN AN AREA THAT IS UNOBSTRUCTED BY TREES, ROOF OVERHANGS, OR ANY OTHER OVERHEAD OBJECT. THE SENSOR SHALL NOT BE PLACED WITHIN THE SPRAY ZONE OF ANY SPRINKLER HEAD, INCLUDING OFF-SITE IRRIGATION. CONTRACTOR SHALL LOCATE SENSOR WITHIN CLOSE PROXIMITY TO THE IRRIGATION CONTROLLER.
31. CONTRACTOR SHALL PERFORM HYDRO-TESTING OF MAIN LINES.

    HYDRO-TESTING TO BE PERFORMED AS LISTED:

    THE CONTRACTOR SHALL NOTIFY THE OWNER'S REPRESENTATIVE FORTY-EIGHT (48) HOURS IN ADVANCE OF TESTING. PRIOR TO BACKFILLING, CONTRACTOR SHALL FILL PIPING WITH WATER, IN THE PRESENCE OF THE OWNER'S REPRESENTATIVE, TAKING CARE TO PURGE THE AIR FROM IT. A SMALL, HIGH PRESSURE PUMP OR OTHER MEANS OF MAINTAINING A CONTINUOUS WATER SUPPLY SHALL BE CONNECTED TO THE PIPING AND SET SO AS TO MAINTAIN 125 PSI FOR TWO (2) HOURS WITHOUT INTERRUPTION. CONTRACTOR SHALL MAKE ANY NECESSARY CORRECTIONS AND RETEST THE SYSTEM UNTIL IT PASSES HYDRO-TEST.
32. ALL WIRING FOR CONNECTION OF THE VALVES TO THE CONTROLLER SHALL FOLLOW MANUFACTURERS SPECIFICATIONS.
33. ALL CONTROL WIRE SHALL BE SLEEVED UNDER PAVEMENT. SLEEVING SIZED SO THAT WIRES CAN BE PULLED FREELY FROM EITHER SIDE OF SLEEVE.
34. CONTRACTOR TO MINIMIZE IRRIGATION OVERTHROW TO IMPERVIOUS AND NATURAL AREAS THROUGH FIELD ADJUSTMENTS TO INDIVIDUAL HEADS.
35. ALL UNIMPROVED AREAS DISTURBED DURING CONSTRUCTION SHALL BE RESTORED TO THEIR ORIGINAL CONDITION BY THE CONTRACTOR.
36. ALL PLANT MATERIAL IN TREE HOLDING AREAS SHALL BE MANUALLY WATERED/IRRIGATED TO KEEP MOIST UNTIL PLANTED.

EXISTING IRRIGATION NOTES

1. CONTRACTOR TO CONFIRM EXISTING SYSTEM SOURCE IS FULLY OPERATIONAL.
2. CONTRACTOR TO TEST EXISTING VALVE WIRES TO PROPOSED IRRIGATION CONTROLLER. ADDITIONAL WIRES WILL NEED TO BE RUN TO ACCOMMODATE ADDITIONAL VALVES AND MASTER VALVE / FLOW SENSOR.
3. CAP EXISTING LINES WHERE NECESSARY FOR CONSTRUCTION.
4. MAINLINE APPROXIMATE LOCATION AS SHOWN. CONTRACTOR TO LOCATE EXACT LOCATION AND UTILIZE. PROTECT MAINLINE IN PLACE DURING CONSTRUCTION. CAP EXISTING MAINLINE AND LEAVE IN PLACE WHERE NECESSARY.
5. DO NOT TRENCH UNDER EXISTING TREES. HAND TRENCH LATERALLY TO INSTALL HEADS, DRIP, ETC.
6. ALL EXISTING IRRIGATION SHOWN TO REMAIN SHOULD BE PROTECTED DURING CONSTRUCTION. CONTRACTOR TO CONFIRM 100 PERCENT COVERAGE ON EXISTING AND MODIFY IF NEEDED.

NOTE: CONTRACTOR TO REMOVE ALL EXISTING COMPONENTS IN THE SOUTH LAWNS TURF AREAS. EXAMPLES OF COMPONENTS ARE EXISTING VALVES, VALVE BOXES, AND HEADS.

EXISTING PIPE WILL BE ABANDONED IN PLACE.

CRITICAL ANALYSIS

Generated:	2025-05-13 12:12
P.O.C. NUMBER: 01	
Water Source Information:	Existing water meter to remain.
Contractor to replace existing backflow with RPZ per Plans.	
55 psi confirmed at existing backflow	
FLOW AVAILABLE	
Water Meter Size:	1"
Flow Available	20.14 GPM
PRESSURE AVAILABLE	
Static Pressure at POC:	55 PSI
Elevation Change:	4.00 ft
Service Line Size:	1"
Length of Service Line:	20 ft
Booster Pump pressure provided:	30 PSI
Pressure Available:	81 PSI
DESIGN ANALYSIS	
Maximum Station Flow:	20.05 GPM
Flow Available at POC:	20.14 GPM
Residual Flow Available:	0.09 GPM
Critical Station:	9
Design Pressure:	45 PSI
Friction Loss:	1.06 PSI
Fittings Loss:	0.11 PSI
Elevation Loss:	0 PSI
Loss through Valve:	3.27 PSI
Pressure Req. at Critical Station:	49.4 PSI
Loss for Fittings:	0.97 PSI
Loss for Main Line:	9.68 PSI
Loss for POC to Valve Elevation:	0 PSI
Loss for Backflow:	11.1 PSI
Loss for Master Valve:	0.36 PSI
Loss for Water Meter:	2.22 PSI
Critical Station Pressure at POC:	73.8 PSI
Pressure Available:	81 PSI
Residual Pressure Available:	7.24 PSI

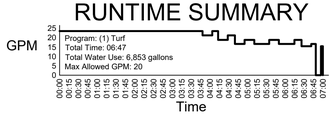
VALVE SCHEDULE

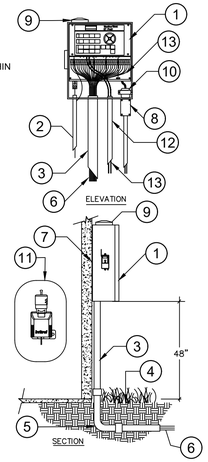
NUMBER	MODEL	SIZE	TYPE	GPM	HEADS	PIPE	DESIGN PSI	FRICTION LOSS	VALVE LOSS	PSI @ POC
1	IRRITROL 2400T	1"	TURF ROTOR	14.2	4	119.3	45	0.76	3.13	64.3
2	IRRITROL 2400T	1"	TURF ROTOR	20.05	5	156.4	45	1.01	3.27	70.7
3	IRRITROL 2400T	1"	TURF ROTOR	20.05	5	169.7	45	1.18	3.27	70.8
4	IRRITROL 2400T	1"	TURF ROTOR	20.05	5	167.1	45	1.14	3.27	70.8
5	IRRITROL 2400T	1"	TURF ROTOR	20.05	5	152.6	45	1	3.27	71.2
6	IRRITROL 2400T	1"	TURF ROTOR	14.2	4	128.9	45	0.81	3.13	65.7
7	IRRITROL 2400T	1"	TURF ROTOR	13.18	4	119.1	45	0.66	3.34	65.4
8	IRRITROL 2400T	1"	TURF ROTOR	20.05	5	145.4	45	1	3.27	73.6
9	IRRITROL 2400T	1"	TURF ROTOR	20.05	5	162.4	45	1.17	3.27	73.8
10	IRRITROL 2400T	1"	TURF ROTOR	14.2	4	221.2	45	1.16	3.13	67.1
11	IRRITROL 2400T	1"	TURF ROTOR	16.04	4	126.9	45	1.02	3.03	68.7
12	IRRITROL 2400T	1"	TURF ROTOR	18.11	5	156.9	45	1.02	3.15	71.6

WATERING SCHEDULE

NUMBER	MODEL	TYPE	PRECIP	SUN	MON	TUE	WED	THU	FRI	SAT	IN./WEEK	MIN./WEEK	GAL./WEEK	GAL./DAY
1	IRRITROL 2400T	TURF ROTOR	0.68 in/h	52 min		52 min		52 min			1.75	154	2,187	729
2	IRRITROL 2400T	TURF ROTOR	0.61 in/h	57 min		57 min		57 min			1.75	171	3,429	1,143
3	IRRITROL 2400T	TURF ROTOR	0.28 in/h	126 min		126 min		126 min			1.75	377	7,559	2,520
4	IRRITROL 2400T	TURF ROTOR	0.28 in/h	126 min		126 min		126 min			1.75	378	7,579	2,526
5	IRRITROL 2400T	TURF ROTOR	0.6 in/h	59 min		59 min		59 min			1.75	176	3,529	1,176
6	IRRITROL 2400T	TURF ROTOR	0.64 in/h	55 min		55 min		55 min			1.75	165	2,343	781
7	IRRITROL 2400T	TURF ROTOR	0.83 in/h	42 min		42 min		42 min			1.75	126	1,661	554
8	IRRITROL 2400T	TURF ROTOR	0.65 in/h	54 min		54 min		54 min			1.75	161	3,228	1,076
9	IRRITROL 2400T	TURF ROTOR	0.3 in/h	117 min		117 min		117 min			1.75	350	7,018	2,339
10	IRRITROL 2400T	TURF ROTOR	0.65 in/h	54 min		54 min		54 min			1.75	162	2,300	767
11	IRRITROL 2400T	TURF ROTOR	0.27 in/h	130 min		130 min		130 min			1.75	389	6,240	2,080
12	IRRITROL 2400T	TURF ROTOR	0.59 in/h	60 min		60 min		60 min			1.75	179	3,242	1,081
		TOTALS:		932		932		932				2,788	50,313	16,771

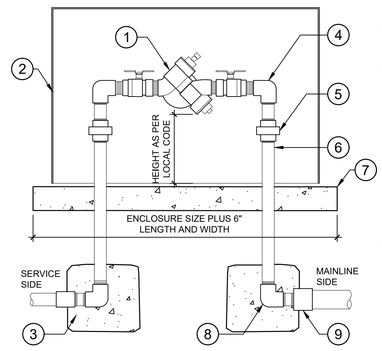
RUNTIME SUMMARY





1. HYDRO POINT WTOXR-C-18-CWM SERIES CONTROLLER WITHIN VANDAL RESISTANT POWDER COATED COLD ROLLED STEEL WALL MOUNT ENCLOSURE ASSEMBLY W/ KEY- LOCK. SEE SPECS FOR SIZE AND FINISH.
2. #6 COPPER GROUND WIRE IN CONDUIT. CONNECT TO BUILDING GROUND AS REQUIRED BY PER ASIC GUIDELINES.
3. 1-1/2" PVC SCH. 40 CONTROL WIRE CONDUIT.
4. FINISH GRADE IN LANDSCAPE AREA.
5. 1-1/2" PVC LONG SWEEP ELL AND CONDUIT . - SIZE AS REQUIRED.
6. LOW VOLTAGE CONTROL WIRING FROM AUTOMATIC CONTROLLER TO ELECTRIC CONTROL VALVES.
7. WALL SURFACE. ANCHOR CONTROLLER TO WALL PER MANUFACTURER'S INSTRUCTIONS.
8. "J" BOX FOR 120 VOLT CONNECTION TO CONTROLLER TRANSFORMER.
9. LOW PROFILE ANTENNA.
10. CONTROLLER TRANSFORMER.
11. RAIN SENSOR PER SPECS (WIRELESS SHOWN)
12. 3/4" PVC SWEEP ELL AND CONDUIT FOR FLOW SENSOR CABLE AND MASTER VALVE WIRES.
13. FLOW SENSOR CABLE AND MASTER VALVE WIRES PER SPECIFICATIONS.

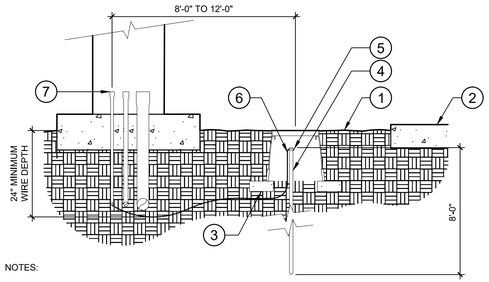
1 HYDRO POINT WTOXR-C-18-CWM



1. REDUCED PRESSURE BACKFLOW DEVICE AS SPECIFIED
2. BACKFLOW ENCLOSURE AS SPECIFIED
3. CONCRETE THRUST BLOCKS, 90 LBS OF CONCRETE EACH
4. GALVANIZED NIPPLES AND ELL AS REQUIRED
5. GALVANIZED UNIONS AT EACH SIDE
6. GALVANIZED RISERS
7. 4" THICK MIN. CONCRETE PAD
8. GALVANIZED ELL AND NIPPLE, TYPICAL
9. PVC COUPLER OR REDUCER AS REQUIRED, TYPICAL

2 REDUCED PRESSURE BACKFLOW W/ ENCLOSURE

1" = 1'-0" 067-606-016-09



NOTES:

1. ALL GROUNDING REQUIREMENTS FOR CONTROLLERS SHALL CONFORM TO LOCAL ELECTRIC CODES.
2. GROUNDING ROD SHALL NOT BE LOCATED IN THE SAME TRENCH AS IRRIGATION MAINLINES OR LATERAL LINES.
3. VALVE BOX SHALL BE WRAPPED WITH A MINIMUM 3 MIL THICK PLASTIC AND SECURED TO THE VALVE BOX USING DUCT TAPE OR ELECTRICAL TAPE.
4. INSTALL GROUNDING ROD PER THE CONTROLLER MANUFACTURER'S SPECIFICATIONS AND RECOMMENDATIONS.

1. FINISH GRADE
2. PAVEMENT
3. THREE (3) 4" x 8" BRICKS
4. 5/8" x 8'-0" COPPER GROUNDING ROD
5. GROUNDING ROD CLAMP
6. #6 AWG BARE COPPER WIRE
7. 1/2" PVC ELECTRICAL CONDUIT AND SWEEP FOR EARTH GROUND

3 GROUNDING ROD

1" = 1'-0" 067-606-016-01

DETAIL - N.T.S.

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PAUL REED Irrigation Designer CID 84393

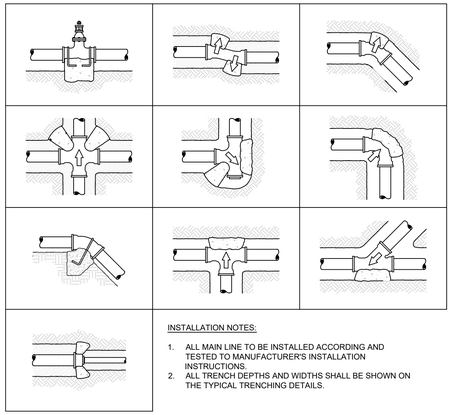
KHA PROJECT 067606016			
DATE 7/1/2025	SCALE AS SHOWN	DESIGNED BY PMR	
		DRAWN BY PMR	
		CHECKED BY AMC	

IRRIGATION SCHEDULES

ARAPAHOE COUNTY FAIRGROUNDS
25690 E QUINCY AVE
AURORA, CO 80016

SHEET NUMBER
IR 2.0

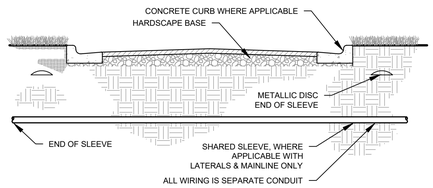
Plotted By:Reed, Paul Sheet Set:Kha Layout:IR 2.0 IRRIGATION NOTES July 11, 2025 03:23:53pm K:\DEN_LA\067606016 — Arapahoe County Fairgrounds\CAD\PlanSheets\067606016_IRR.dwg
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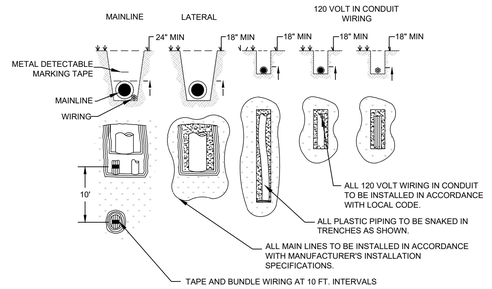
INSTALLATION NOTES:

1. ALL MAIN LINE TO BE INSTALLED ACCORDING AND TESTED TO MANUFACTURER'S INSTALLATION INSTRUCTIONS.
2. ALL TRENCH DEPTHS AND WIDTHS SHALL BE SHOWN ON THE TYPICAL TRENCHING DETAILS.

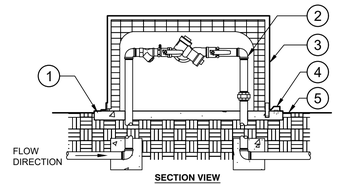
1 TYPICAL THRUST BLOCK
IR2.0 SECTION — NTS



2 TYPICAL SLEEVING
IR2.1 SECTION — NTS



3 TYPICAL TRENCHING
IR2.0 SECTION — NTS

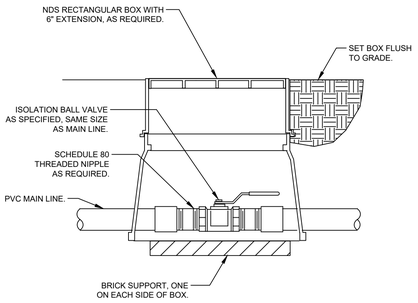


1. 4" THICK CONCRETE FOOTING
2. 1" ABOVE FINISHED GRADE BACKFLOW PREVENTION DEVICE (SEE IRRIGATION PLAN FOR MAKE AND MODEL)
3. BACKFLOW CAGE (SEE IRRIGATION LEGEND FOR MAKE AND MODEL)
4. LOCK BOX
5. FINISHED GRADE

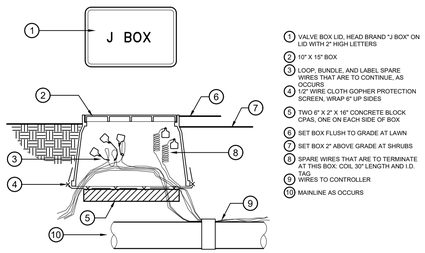
NOTES:

1. INSTALL BACKFLOW CAGE PER MANUFACTURER'S SPECIFICATIONS AND RECOMMENDATIONS.
2. SEE BACKFLOW PREVENTION DEVICE DETAIL FOR REFERENCE.
3. LOCK BOX SHALL BE LOCATED ABOVE CONCRETE FOOTING
4. CONTRACTOR SHALL PROVIDE A LOCK AS APPROVED BY THE OWNER'S REPRESENTATIVE.

4 BACKFLOW CAGE
3/8" = 1'-0" — 067-606-016-02

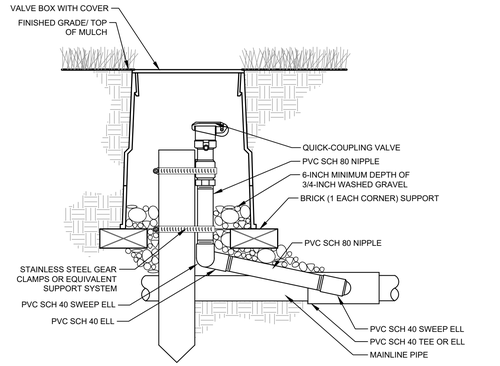


5 BRASS BALL ISOLATION VALVE
IR-2.1 DETAIL SUBTITLE — NTS



1. VALVE BOX LID, HEAD BRAND "J BOX" ON LID WITH 2" HIGH LETTERS
2. 10" X 15" BOX
3. LOOP, BUNDLE, AND LABEL SPARE WIRES THAT ARE TO CONTINUE, AS OCCURS
4. 1/2" WIRE CLOTH GOPHER PROTECTION SCREEN, WRAP 6" UP SIDES
5. TWO 6" X 2" X 16" CONCRETE BLOCK CPAS, ONE ON EACH SIDE OF BOX
6. SET BOX FLUSH TO GRADE AT LAWN
7. SET BOX 2" ABOVE GRADE AT SHRUBS
8. SPARE WIRES THAT ARE TO TERMINATE AT THIS BOX: COIL 30" LENGTH AND I.D. TAG
9. WIRES TO CONTROLLER
10. MAINLINE AS OCCURS

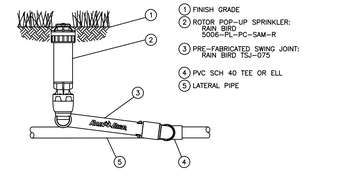
6 WIRE BUNDLE JUNCTION BOX
1" = 1'-0" — 067-606-016-16



NOTE:

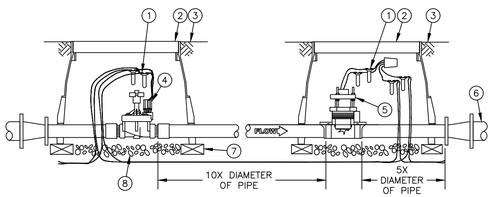
FURNISH FITTINGS AND PIPING NOMINALLY SIZED IDENTICAL TO NOMINAL QUICK COUPLING VALVE INLET SIZE.

7 TYPICAL QUICK COUPLER
IR2.0 SECTION — NTS



1. FINISH GRADE
2. ROTOR POP-UP SPRINKLER: RAIN BIRD 5006-PL-PC-SAM-R
3. PRE-FABRICATED SWING JOINT: RAIN BIRD TSJ-075
4. PVC SCH 40 TEE OR ELL
5. LATERAL PIPE

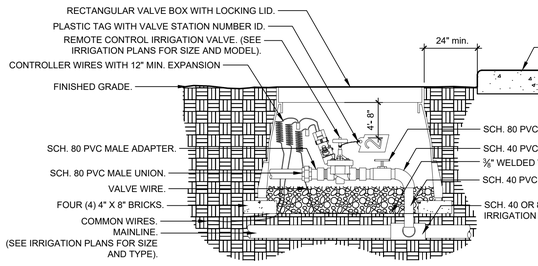
8 PRESSURE REGULATED ROTOR
1" = 1'-0" — 196-008-000-45



1. WATERPROOF SPLICE CONNECTION:
2. VALVE BOX WITH COVER:
3. FINISH GRADE/TOP OF MULCH
4. NORMALLY CLOSED MASTER VALVE:
5. FLOW SENSOR
6. PVC MAINLINE PIPE
7. BRICK (1 OF 4)
8. 3-INCH MINIMUM DEPTH OF 3/4-INCH WASHED GRAVEL

NOTE:
1. COIL AN ADDITIONAL 3-FEET OF CABLE IN VALVE BOX FOR EASY SERVICING.

9 MASTER VALVE AND FLOW SENSOR
NTS — 196-008-000-32



NOTE:
1- LOCATE VALVE BOX WITHIN 24" OF PAVEMENT EDGE IN PLANTING AREA WHERE EASILY ACCESSIBLE WHENEVER POSSIBLE.

2- COMMON WIRE AND CONTROLLER WIRE SHALL BE DIRECT BURIAL 14 AWG OR LARGER. COLOR: COMMON (WHITE), CONTROLLER WIRE FOR TURF (BLUE), AND CONTROLLER WIRE FOR SHRUBS (RED). (SEE SPECIFICATIONS FOR 2-WIRE CONTROLLERS).

3- ALL WIRE RUNS SHALL BE CONTINUOUS WITHOUT ANY SPLICES UNLESS APPROVED BY THE OWNER'S REPRESENTATIVE. SEE SPLICE BOX DETAIL. WIRE CONNECTIONS SHALL BE MADE USING DBR/Y-6 CONNECTORS OR APPROVED EQUAL.

4-VALVE BOX SHALL BE WRAPPED WITH MIN. 3 MIL THICK PLASTIC AND SECURE IT USING DUCT TAPE OR ELECTRICAL TAPE.

5- MAINLINES 4" OR LARGER SHALL USE SADDLES AT THE CONNECTIONS POINTS TO THE IRRIGATION VALVE. (SEE SPECIFICATIONS FOR IRRIGATIONS SADDLES).

6- ALL SCH. 80 PVC TO SCH. 40 PVC THREADED CONNECTIONS SHALL BE MADE USING TEFLON TAPE.

7- VALVE BOXES SHALL BE LOCATED IN PLANTING AREAS.

10 REMOTE CONTROL IRRIGATION VALVE
1 1/2" = 1'-0" — 067-606-016-10

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No.	REVISIONS	DATE	BY

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PHONE: 206-607-2600
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PAUL REED — Professional Irrigation Designer — CID — IRRIGATION ASSOCIATION — 84393

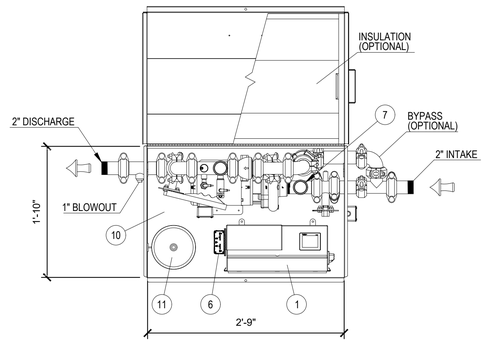
KHA PROJECT	067606016
DATE	7/11/2025
SCALE	AS SHOWN
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IRRIGATION DETAILS

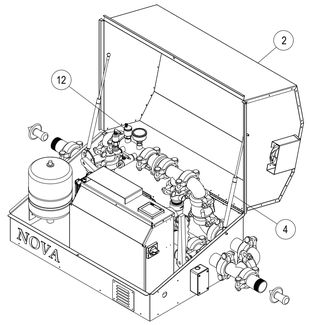
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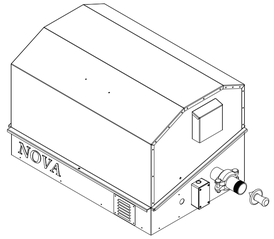
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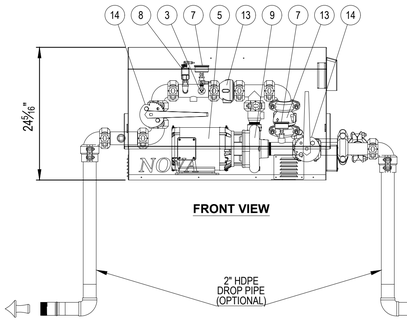
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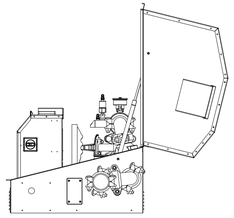
PLAN VIEW



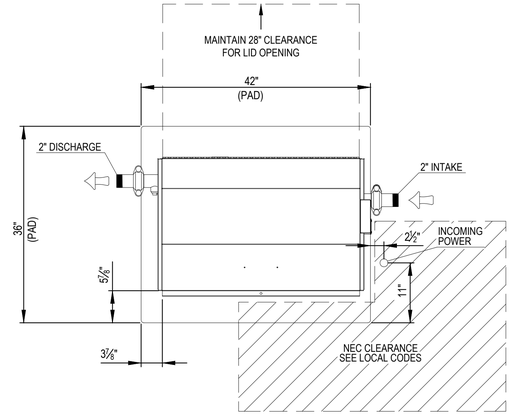




FRONT VIEW



SIDE VIEW



RECOMMENDED PAD LAYOUT

DESIGN SPECIFICATIONS		
DESIGN FLOW RATE:	25 GPM @ 30 PSI BOOST	
POWER:	208 VOLT / 3 PHASE	
MODEL #:	CB##V1C002X00025-040XXXX203ONS-2-SS0	
ITEM NO.	DESCRIPTION	QTY.
1	CS CASE	1
2	ENCLOSURE, NOVA	1
3	FLOW METER/FLOW SWITCH	1
4	GAS STRUT	2
5	MOTOR, 3600 RPM, ODP	1
6	MODEM	1
7	PRESSURE GAUGE	2
8	PRESSURE TRANSDUCER	1
9	PUMP, CENTRIFUGAL	1
10	SKID, POLY	1
11	TANK, PRESSURE	1
12	VALVE, AIR RELIEF, FV-4	1
13	VALVE, CHECK, GROOVED	1
14	VALVE, BUTTERFLY, GROOVE, LEVER	2

MATERIAL PIPING: SKID:		TITLE:	PI SERIES: NOVA-SERIES CENTRIFUGAL BOOSTER PUMP STATION LEFT DISCHARGE, W/BYPASS, 0-90 GPM				
UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN INCHES TOLERANCES: FRACTIONAL: ±1/8" ONE PLACE DECIMAL: ±.10 TWO PLACE DECIMAL: ±.05 DO NOT SCALE DRAWING		PROJECT:	- -				
		PART NO:	PRECISION PUMPING SYSTEMS (208) 323-5300				
DRAWN: JWR	DATE:	JOB:	SIZE: B	SCALE: NTS	SHEET: 1 OF 1	REV	-

SCOPE OF WORK

Packaged Pump Station supplier shall provide a variable speed END SUCTION CENTRIFUGAL pump station complete with pump, piping, valves, sensors, variable frequency drive (VFD), programmable logic controller (PLC), NEMA 3R rated disconnect switch, and all appurtenances necessary for a complete and functioning pumping system. The pump station, including the pump controls, shall be designed and manufactured within the confines of a UL 508A and UL QCZJ certified facility. The station performance specifications shall be as indicated on the contract drawing.

Technical Service and Support. The manufacturer shall provide access to 24/7 technical phone support with a factory certified technician during the warranty period. The technician shall have access to all relevant data specific to the pump station, including specifications, submittal, shop drawings, programming, and detailed photos of the system.

Factory Testing. The pump station shall undergo and pass all of the following system performance tests: Hydrostatic testing that meets ANSI/HI specifications and standards; Flow testing that meets ANSI/HI 14.6 specifications and standards; and Vibration testing that meets ANSI/HI 9.6.4 Vibration Measurement and Allowable Values specifications and standards. The pumping system shall be flow tested as a complete unit, which shall include function testing of pumps, motors, instrumentation, appurtenances, and control system. The results of all tests shall be available to the owner.

PIPING, VALVES, PRESSURE TANK, SKID BASE, & STATION ENCLOSURE

Piping. The station suction and discharge piping shall be 304 STAINLESS STEEL, Schedule 10, assembled with grooved connections. Threaded connections between the main piping sections other than at the pump volute shall not be acceptable.

Valves. Butterfly style isolation valves--with grooved connections--shall be included on station suction and discharge piping. A non-slam check valve shall be included on the discharge of each pump. An air release valve shall be included, located immediately after the pump discharge.

Pressure Tank. The station shall include a 2 gallon diaphragm type pressure tank to stabilize system pressure and prevent the pump from short-cycling.

Station Enclosure & Base. The station shall be enclosed in a powder coated marine grade aluminum enclosure. The enclosure shall include a ventilation fan appropriately sized to adequately cool the enclosed equipment. The station base shall be constructed of 1" thick rigid HDPE.

INSTRUMENTATION

Pressure Gauges shall have a 304 stainless steel case and bezel. Gauges shall have a 2.5" diameter and be liquid filled.

Pressure Transmitter shall be constructed of stainless steel and rated for the pump station discharge pressure.

Flow Switch. The station shall include a stainless steel flow sensor to prevent the system from operating during a no-flow condition.

Flow Sensor. Where indicated, the pump station shall include anon-magnetic insertion type flow sensor.

PUMP SYSTEM CONTROLLER

Control Panel Manufacturing & Testing. The pump control panel shall be manufactured and listed by a UL508A Panel Shop. The panel shall be UL labeled as an "Enclosed Industrial Control Panel". The pump control panel shall be completely manufactured, tested and programmed prior to delivery to the job site.

Controls Enclosure. All control equipment, including the VFD and PLC shall be housed in a UL Type 1 marine grade aluminum controls enclosure. The enclosure shall provide a degree of protection against spraying water. It shall not be acceptable for the VFD or PLC to be mounted in the pumps station enclosure, but not inside a controls enclosure.

Cooling System. The control panel cooling system shall be appropriately sized for the ambient conditions. The cooling system shall not allow dust, insects or rodents inside the pump control panel.

Main Disconnect. A NEMA 3R rated non-fused disconnect switch shall be mounted on the outside of the pump station enclosure and shall be capable of isolating all power to the pump station.

Variable Frequency Drive (VFD). The VFD shall be appropriately sized to meet the FLA (full load amps) required by the pump motor, as stated on the motor nameplate. The VFD shall be manufactured by ABB Industrial Systems, Mitsubishi, or approved equal. Initial start-up and calibration shall be performed by a factory certified technician, which shall extend the warranty on the control panel to a total of three (3) years.

Programmable Logic Controller (PLC). The PLC shall be fully programmed prior to pump panel installation. The technician installing and programming the PLC is to be factory trained and certified by the PLC manufacturer. The PLC programming shall be non-proprietary, with the complete station programming shall be made available to the owner via a USB drive included with the station control panel.

Operator Interface. The controller shall be equipped with a 3.5" LCD color touchscreen. The operator interface shall allow the user to make adjustments to the PLC program locally without requiring any additional equipment such as a laptop computer. A VFD control keypad is not an acceptable substitution for the digital operator interface. The PLC shall have an Ethernet port to enable remote access.

Control Functions

1. User settable Local or Remote control.
2. Pre-Programmed Start-Up Routines to limit and/or delay starting and acceleration of the pump to eliminate excessive velocity and pressure. It shall also include initial start-up, mainline fill, power outage and automatic re-starts.
3. System Pressure Setpoint, with user adjustable discharge setpoint and alarm settings.
4. Pump Sleep Settings, with two threshold modes: Sleep by Flow or Sleep by Frequency.
5. System Protection Settings, including fault and warning parameters for low flow, high flow, low pressure, high pressure, restart trials and restart delay time.
6. Start-up Assistant, including VFD programming, sensor activation & calibration, operational settings, pressure setpoint, & pump sleep settings.
7. Troubleshooting Assistant for all system alarms and faults.
8. Load Factory Default Settings, User Saved Default Settings.
9. Password Protection, with the ability to create different accounts with access restrictions.

Monitoring Functions

1. Pump operating status, total pump run hours, motor frequency, motor amperage
2. System pressure, flow data
3. Fault Log with time stamps and diagnostic utility.
4. Trend Data, with graphic display of system pressure, flow, motor frequency and amperage. Data shall be exportable to MS Excel.

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No.	REVISIONS	DATE	BY

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PAUL REED Smart Irrigation Specialist
CID
84393

KHA PROJECT 067606016			
DATE 7/1/2025			
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IRRIGATION PUMP STATION

ARAPAHOE COUNTY FAIRGROUNDS
25690 E QUINCY AVE
AURORA, CO 80016

SHEET NUMBER
IR 2.2

Plotted By:Reed, Paul Sheet Set:Kha Layout:IR 2.2 PUMP STATION July 11, 2025 03:24:03pm K:\DEN_LA\067606016 – Arapahoe County Fairgrounds\CAD\PlanSheets\067606016_IRR.dwg
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